

Jose Bande Serrano

Full-stack Software Engineer

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Professional Summary

Full-stack Software Engineer with 10+ years of experience building production applications in collaborative engineering environments. Offers strong evidence in Codex, OpenAI, AI Agents, AI-Assisted Development, Python, and JavaScript, with additional hands-on background in production software delivery. Experience spans APIs, data-intensive systems, automation platforms, and cloud-based application delivery.

Selected Skills

Codex, JavaScript, Python, SQL, TypeScript, REST APIs, AI Agents, AI-Assisted Development, Azure SQL, Containerization, FastAPI, OpenAI, PostgreSQL, SQL Server, Tailwind CSS, Web Scraping, .NET, Application Modernization, ASP.NET, Auth0, Authentication, Azure, Azure App Service, Azure DevOps

Professional Experience

NBCUniversal - Full-Stack Engineer, Compliance Automation

2024-09 - present | New York, NY

Built and delivered a production compliance automation platform combining a high-performance Next.js SPA, backend workflows, ETLs, and reporting systems.

- Implemented AI bots integrated with Microsoft Teams.
- Designed and implemented tens of Dataverse tables to support the application's data model.
- Created dozens of Power Automate flows and dataflows to support compliance workflows and data movement.
- Developed a React application embedded inside a model-driven Power Apps application.
- Used Codex to accelerate software development and support project management work.

Enerex - AI Development Consultant

2023-11 - present

Architected frontend and AI service layers for task-based automation systems and cross-product integrations.

- Built production applications with Codex as a primary development tool, accelerating implementation across frontend, AI service, and delivery work.
- Built an AI-powered Outlook add-in to automate contract processing workflows for energy retail market users.
- Designed the add-in to open directly from contract emails, analyze attached contracts with an agent carrying deep business knowledge, and return standardized extracted data.
- Provisioned and deployed Azure Functions, web applications, databases, and Storage Queues to support Enerex products.
- Implemented deployment pipelines for application delivery and environment updates.

InteroCloud - Senior Full-Stack Developer

2022-11 - 2023-11

Contributed to a medical-practice CRM with Next.js SSR, Playwright automation, SEO improvements, and Python migration tooling.

- Worked in Linux environments while developing and supporting production healthcare software systems.
- Developed and maintained production systems across PHP, Vue, and JavaScript using Bootstrap, Docker Compose, Nuxt.js, and Vue.js.
- Applied REST APIs, Containerization, and Database Design practices in production systems.
- Delivered frontend features and API-integrated workflows in production systems.

First Due - Senior Software Engineer

2017-07 - 2019-11

Built large-scale web crawlers and ETL pipelines with Python and event-driven backend systems.

- Designed and maintained PostgreSQL databases supporting large-scale data collection workloads.
- Implemented tens of Python web crawlers with the Scrapy framework to extract critical company data from public websites.
- Built crawler orchestration and monitoring capabilities using Django, FastAPI, and RabbitMQ.
- Worked in Linux environments while building and operating Python-based data extraction and ETL systems.
- Architected a system to manage and control web crawlers, centralizing execution control and operational monitoring in a single UI.

Liberty Power - .NET/SQL Consultant

2016-05 - 2017-06

Modernized legacy .NET applications and improved database quality and performance with SQL work.

- Implemented and ran SQL queries to investigate, validate, and repair data issues.
- Migrated projects from Visual Basic to ASP.NET as part of legacy application modernization work.
- Implemented numerous stored procedures to support application behavior and database operations.
- Performed extensive database data consistency reviews and corrections across business-critical datasets.

Projects

AI Assisted Invoice Processor

An application to help finance teams collect invoices, extract structured data, review exceptions, and move approved records into downstream reporting without juggling disconnected inboxes, sheets, and shared folders.

- Built the backend with Node.js, Express.js, and Prisma over SQL Server hosted in Azure.
- Built the frontend with React.js, Next.js, Tailwind CSS, Lucide React, and TanStack to support real-time, data-driven invoice operations.
- Integrated Nylas and Google services to enable controlled access to user email inboxes for invoice ingestion.
- Implemented AI-powered invoice analysis using OpenAI.

Education

Bachelor's Degree, Engineer in Electronics

Ciudad Universitaria Jose A. Echeverria (CUJAE) | 2004 - 2009